Boonyaporn Preechasuth

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Education

King Mongkut's University of Technology Thonburi, Bangkok, Thailand 2023 – Present
B.Eng. Robotics and Automation Engineering, Institute of Field Robotics (FIBO)

- Current GPAX: 3.91 / 4.00 (Cumulative over 6 semesters)
- **Coursework**: Deep Reinforcement Learning, Machine Learning, Autonomous Mobile Robots, Control Engineering, Kinematics, Robotics Modeling, Embedded Systems, and Microcontroller Interface
- Expected Graduation: May 2027

Experience

Research Intern Jun 2026 – Nov 2026
BRAIN Lab, School of Information Science and Technology (IST), VISTEC

- Undertaking a six-month Work-integrated Learning (WiL) research placement.
- Researching sim-to-real transfer and inverse reinforcement learning for legged locomotion.

Teaching Assistant (Super TA) Aug 2024 – Present
Institute of Field Robotics (FIBO)

- **Computer Programming (Python):** Taught programming fundamentals to first-year students and designed weekly coding challenges assessing algorithmic thinking.
- **Statics and Strength of Materials:** Led the course term project for first-year students, graded homework assignments, and tracked individual student progress.
- **Robotic Modeling and Experimentation:** Ran MATLAB/Simulink laboratory sessions on sensors, actuators, and system modeling for second-year students.
- Awarded the **Super Teaching Assistant Scholarship**, providing full tuition support.

Publication

Autonomous Cleaning of Solar Panels in Floating Photovoltaic Systems Dec 2025
Keerati Ubolmart, **Boonyaporn Preechasuth**, Natthapatch Lapsittiwong, et al.
2025 5th International Conference on Robotics, Automation, and Artificial Intelligence (RAAI 2025)
DOI: 10.1109/RAAI67517.2025.11423055

Research and Projects

AIRL for Stick Insect Locomotion Jul 2026 – Present

- Applying Adversarial Inverse Reinforcement Learning (AIRL) to a stick insect locomotion task.
- Training and evaluating locomotion policies in CoppeliaSim.

Sim-to-Real Transfer (Unitree B1) Jun 2026 – Jul 2026

- Trained locomotion policies in Isaac Lab, validated them in MuJoCo, then deployed to a real B1.
- Transferred policies to hardware using PACE (Precise Adaptation through Continuous Evolution).
- Identified motor and actuator parameters from hardware to match simulated drivetrain response.
- Applied domain randomization over the identified parameters to close the reality gap.

Continual Reinforcement Learning for Quadruped Locomotion Feb 2026 – May 2026

- Surveyed and compared continual learning methods for mitigating catastrophic forgetting.
- Applied the selected methods to reinforcement learning locomotion on a Unitree Go2.
- Benchmarked policy retention and transfer across sequential locomotion tasks.

AMP-Anchored Continual Reinforcement Learning | *GitHub* Apr 2026 – May 2026

- Extended the continual reinforcement learning work with Adversarial Motion Priors (AMP).
- Anchored policy behavior to reference motions to preserve gait quality across tasks.

- Reduced degradation of previously learned skills when adapting to new locomotion tasks.

F1TENTH Autonomous Racing Vehicle | *GitHub/CARVER-F1TENTH* Mar 2026 – May 2026

- Developed an autonomous 1:10-scale F1TENTH vehicle that tracks an optimized raceline.
- Implemented collision-free path planning to follow the raceline through a cluttered track.
- Improved Pure Pursuit-based obstacle avoidance to steer around obstacles and rejoin the line.

Autonomous Solar Panel Cleaning Robot Jun 2025 – May 2026

- Developed an autonomous dual-track robot for cleaning floating photovoltaic systems, funded by EGAT.
- Implemented a modified Stanley controller to hold parallel alignment and offset from panel edges.
- Engineered real-time edge detection from dual RPLIDAR S3 sensors with point-cloud segmentation.
- Integrated a hierarchical ROS2 and micro-ROS stack on an STM32G474RE for actuator control.
- Designed an adaptive brush module using dual load cells to regulate force via pressure feedback.

Sim-to-Sim Transfer (Unitree Go2) | *GitHub/Sim2Sim* Feb 2026

- Trained locomotion policies in Isaac Lab and tested transfer in MuJoCo.
- Applied domain randomization and an actuator network to model drivetrain dynamics.

CARVER - Autonomous Driving Car | *GitHub/CARVER* Aug 2025 – Dec 2025

- Developed an autonomous Ackermann-steering vehicle with path-following capabilities.
- Integrated 3D LiDAR and IMU sensors for localization and navigation along pre-planned paths.
- Evaluated Pure Pursuit, Stanley, and combined controllers by comparing cross-track and heading errors.
- Utilized ROS2 and micro-ROS with ODrive for motor control and real-time visualization in RViz2.

Rotary Inverted Pendulum (RIP) | *GitHub/RIP* Aug 2025 – Dec 2025

- Performed kinematic analysis and modeling to derive system dynamics and transformation matrices.
- Developed embedded control on a NUCLEO-G474RE with Colors/LCD Shields and Bluetooth.
- Implemented high-speed SPI data logging to a FAT32 file system for offline analysis.
- Evaluated stability through real-time joint-state visualization in ROS2 and RViz2.

FloodNav - Intelligent Navigation System Sep 2025 – Nov 2025

- Developed a path-planning algorithm for rescue routes from flood depth and road accessibility.
- Implemented ConvNeXt–UPerNet–DGCN segmentation for road extraction from satellite imagery.
- Integrated satellite-derived road maps with crowd-sourced field data for response awareness.
- **Awards:** Top 5, GeoHackathon THEOS-2; Top 10, Ford Smart Mobility Challenge.

TPA Super Dunk, ABU Robocon 2025 Apr 2025 – May 2025

- Compared OpenCV stereo depth, Unitree 4D LiDAR, and Intel RealSense D400 for target precision.
- Implemented the final vision pipeline on Intel RealSense with YOLOv11 basketball tracking.
- **Award:** National Finalist (Top 24), ABU Robocon Thailand.

2-DOF Vertical Plotter Jan 2025 – May 2025

- Led the project and directed the control systems division.
- Performed motor identification and developed a cascade PID controller with S-curve trajectory planning.
- Engineered a Kalman Filter for state estimation to ensure high-precision positioning.

Automated Soft-shell Crab Harvesting System Aug 2024 – Nov 2024

- Engineered an automated system to identify and harvest soft-shell crabs during the molting phase.
- Developed an infrared machine vision system for low-light monitoring without disrupting the cycle.
- Evaluated YOLOv8 object detection against instance segmentation for morphological change detection.
- **Award:** Top 10, Ford Innovator Scholarship.

Automatic Squash Ball Launcher Jan 2024 – May 2024

- Led the development of a spring-powered automated launcher with lead-screw positioning.
- Designed mechanical specifications and electrical control circuits to ensure accurate target coverage.
- Engineered a 2D simulation using Pygame to model ballistics before physical fabrication.

Honors and Awards

Super Teaching Assistant Scholarship - Full Tuition Support 2023 – Present

Top 5 Finalist, GeoHackathon THEOS-2 (GISTDA) 2025

Top 10 Finalist, Ford Smart Mobility Challenge 2025

National Finalist (Top 24), TPA Super Dunk - ABU Robocon Thailand 2025

Top 10 Finalist, Ford Innovator Scholarship 2024

Leadership and Activities

Deputy Head of Academic Affairs, FIBO Robocamp XII 2024

- Designed the academic curriculum and final robotics mission for high school participants.
- Served as lead Programming Instructor, also supporting mechanics and electronics teaching.

Academic Affairs Lead, Fibolian XI 2024

- Led academic programming, including the Kick-off Project for incoming robotics students.
- Coordinated camp operations and cross-departmental integration as an Executive Team member.

Executive Team, FIBO Demo Day 2023 2023

- Supported planning and execution of project demonstrations and large-scale event operations.

Skills

Reinforcement Learning: Deep Reinforcement Learning, Continual Learning, Imitation Learning, Adversarial Motion Priors (AMP), Adversarial Inverse Reinforcement Learning (AIRL), Domain Randomization, Actuator Network Modeling, Sim-to-Sim and Sim-to-Real Transfer

Robotics and Control: ROS2 (Humble), micro-ROS, Kinematics, Trajectory Planning, Path Planning, Cascade Control, Kalman Filter, State Estimation, SLAM, Navigation

Programming and Computer Vision: Python, C/C++, PyTorch, MATLAB/Simulink, Image Processing, Object Detection, Instance Segmentation, Semantic Segmentation

Embedded Systems and Sensors: Microcontrollers (STM32, Arduino, ESP32), Communication Protocols (UART, SPI, I2C, CAN, Modbus, MQTT), LiDAR, Intel RealSense Camera

Simulation and Tools: Isaac Lab, MuJoCo, CoppeliaSim, Gazebo, RViz2, Ubuntu, Docker, Git/GitHub, VS Code, TensorBoard, SolidWorks (Basic), EasyEDA (Basic), Microsoft Office, LaTeX

Languages: Thai (Native), English (IELTS Academic 7.0, Oct 2026)